



L5a: Single Asset Geometric Brownian Motion and NPV

CHEME 5660 Cornell University

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you will be able to:

- **Interpret out-of-sample predictions:** explain how historical mean growth and volatility determine the distribution of later prices.
- **Calculate an NPV target probability:** connect the discounted trade payoff to the probability of exceeding a target at the sale date.
- **Update GBM parameter estimates:** apply exponentially weighted updates for mean growth and volatility, explain the half-life, and use the estimates in a forecast with fixed parameters.

Lectures and Examples

Example: Out-of-Sample Single-Asset GBM

How well do historical estimates describe later prices? Simulate 2025 prices, construct prediction bands, and compare observed coverage across assets.

Example: GBM Net Present Value Trade Rule

How likely is a trade to exceed a target scaled NPV? Calculate the probability, check it at the median, and examine how it changes with the target.

Example: Updating GBM Parameters with EMA

Can recent observations improve the forecasts? Update mean growth and volatility, then compare the resulting forecasts with the frozen model.

Company Profile: Renaissance Technologies

Renaissance Technologies (RenTec) was founded by [Jim Simons \(1938–2024\)](#), a mathematician and former chair of mathematics at Stony Brook University.

The firm uses mathematical and statistical methods to guide investments. Its researchers come from mathematics, physics, computer science, and related fields.

Its **Medallion Fund**, launched in 1988, averaged about 66% annually before fees and 39% after fees over 1988–2018. These are arithmetic averages of annual returns, not compound growth rates. The fund is closed to outside investors; RenTec's other funds use different strategies.

Sources: [RenTec overview](#); [Bradford Cornell's study of Medallion](#).

Renaissance Technologies: Explore Further

Jobs and internships. Browse the [RenTec careers page](#) for research, programming, and systems roles. No internships were listed as of September 19, 2026; check for new openings.

Watch Jim Simons discuss his work:

- [Numberphile: full-length interview](#) ([YouTube channel](#)).
- [TED: The mathematician who cracked Wall Street](#) ([YouTube channel](#)).

Connection to today: Build a model, record its predictions, and compare them with later observations. Our GBM examples introduce this research process; they do not describe RenTec's proprietary strategies.

Concept Review: Single Asset GBM Model

Recall that geometric Brownian motion (GBM) describes share price changes through drift and random fluctuations, both proportional to the current price:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu dt}_{\text{drift}} + \underbrace{\sigma dW(t)}_{\text{random fluctuations}}.$$

Here, $S(t)$ is the share price (dollars per share), with initial price $S(0) = S_0 > 0$, and $dW(t)$ is a Wiener-process increment.

With time measured in years, the constant parameters are:

- **Drift** $\mu \in \mathbb{R}$: the expected proportional price change per unit time (year^{-1}).
- **Volatility** $\sigma > 0$: sets the size of the random fluctuations ($\text{year}^{-1/2}$).

Concept Review: Growth and Volatility

Over an observation interval $\Delta t > 0$ years, the growth rate is:

$$g_j = \frac{1}{\Delta t} \ln \left(\frac{S_{t_j}}{S_{t_{j-1}}} \right), \quad t_j = j\Delta t.$$

Under GBM, its mean and standard deviation are:

$$\underbrace{\mu_g = \mathbb{E}[g_j] = \mu - \frac{\sigma^2}{2}}_{\text{mean growth rate}}, \quad \underbrace{\sqrt{\text{Var}(g_j)} = \frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate standard deviation}}.$$

Drift μ and mean growth rate μ_g both have units of year^{-1} , but differ by $\sigma^2/2$.

From the sample growth-rate standard deviation σ_g , we estimate the volatility parameter as:

$$\hat{\sigma} = \sigma_g \sqrt{\Delta t}.$$

Concept Review: The One-Step Transition

For a grid of N steps, $t_j = j\Delta t$, the exact transition from L4b is:

$$S_{t_j} = S_{t_{j-1}} \exp\left[\mu_g \Delta t + \sigma \sqrt{\Delta t} Z_j\right], \quad j = 1, \dots, N.$$

Here, $Z_1, \dots, Z_N$ are independent standard normal draws, one per step.

For a single horizon $T > 0$, we can sample the terminal price directly:

$$S_T = S_0 \exp\left[\mu_g T + \sigma \sqrt{T} Z\right], \quad Z \sim \mathcal{N}(0, 1).$$

Repeated one-step draws generate a price path. The fixed-horizon form samples the price at the selected time T .

Example: Out-of-Sample Price Predictions

Example: Test predictions against later observations

How well does a model fitted to historical prices describe prices observed afterward?

- Use the mean growth rate and volatility estimated from 2014–2024 data.
- Simulate prices for 2025 and compare prediction bands with observed prices.
- Measure the fraction of dates with observed prices inside each band, first for one firm and then across the dataset.

Next question: what is the probability that a scheduled sale exceeds a target scaled NPV?

GBM Trade Rule

Consider a scheduled trade: buy $n_0 > 0$ shares at S_0 at time 0 and sell at S_T at time $T = N\Delta t$. Assume no dividends or trading costs.

Let g_y be the constant, continuously compounded benchmark growth rate (year^{-1}) corresponding to yield y . The NPV is given by:

$$\text{NPV}(g_y, T) = \underbrace{-n_0 S_0}_{\text{purchase today}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{discounted sale}}.$$

Dividing by the initial investment gives the **scaled NPV**, our discounted fractional return:

$$\rho_T = \frac{\text{NPV}(g_y, T)}{n_0 S_0} = \left(\frac{S_T}{S_0} \right) e^{-g_y T} - 1.$$

We now use GBM to describe the uncertain sale price S_T .

Substitute the GBM Solution

Using the mean growth rate $\mu_g = \mu - \sigma^2/2$, substitute the GBM price into the scaled NPV:

$$\rho_T = \exp[(\mu_g - g_y)T + \sigma\sqrt{T}Z] - 1, \quad Z \sim \mathcal{N}(0, 1).$$

Discounting subtracts the benchmark growth rate g_y from the mean growth rate μ_g , while leaving the random shock unchanged. The quantity $1 + \rho_T$ is lognormal, so:

$$\ln(1 + \rho_T) \sim \mathcal{N}((\mu_g - g_y)T, \sigma^2 T).$$

The scaled NPV is a lognormal variable shifted down by one, with $\rho_T > -1$.

The benchmark affects the return through $g_y T$. When $|g_y|T \ll 1$, discounting is small and $\rho_T \approx S_T/S_0 - 1$. We use the exact discounted expression to calculate the target probability.

Terminal Target Probability

Choose a target scaled NPV $\rho_\star > -1$. Adding one and taking the logarithm preserves the strict target condition:

$$\rho_T > \rho_\star \iff \ln(1 + \rho_T) > \ln(1 + \rho_\star).$$

Substituting the GBM expression gives:

$$(\mu_g - g_y)T + \sigma\sqrt{T} Z > \ln(1 + \rho_\star).$$

Subtracting $(\mu_g - g_y)T$ and dividing by $\sigma\sqrt{T} > 0$ gives a threshold for the standard normal random variable Z :

$$Z > z_\star, \quad \boxed{z_\star = \frac{\ln(1 + \rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}}.$$

Thus $Z > z_\star$ means the trade exceeds its target at the scheduled sale. Selling when a boundary is first reached requires a separate calculation.

The Closed-Form Target Probability

The standard normal cumulative distribution function $\Phi(z_\star)$ gives the probability that $Z \leq z_\star$. Subtracting from one gives the probability that $Z > z_\star$:

$$\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi(z_\star) = 1 - \Phi\left(\frac{\ln(1 + \rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}\right).$$

- For a chosen target, we evaluate $z_\star$ from estimated mean growth and volatility, the holding period, and the benchmark growth rate.
- The formula applies for $T > 0$, $\sigma > 0$, and $\rho_\star > -1$.
- A target of -1 or below is exceeded with probability one because the scaled NPV always exceeds -1 under GBM.

Example: Calculate and verify the NPV target probability

Change the Benchmark

The target scaled NPV corresponds to a required sale price:

$$K = S_0(1 + \rho_\star)e^{g_y T}.$$

With $\rho_\star = 0$, the trade succeeds if it beats the benchmark.

- $g_y = 0$: did the price increase?
- Positive g_y : did it increase enough over the holding period?
- Increasing g_y raises the sale threshold and lowers the probability of exceeding it.

The same prices can be assessed against different assumed growth benchmarks.
Choose the benchmark before examining the outcome.

Example: GBM Net Present Value Trade Rule

Example: Evaluate an NPV-based GBM trade rule

How likely is a stock trade to exceed our target scaled NPV at a scheduled sale?

We calculate this probability using the selected firm's mean growth and volatility, the holding period, and the benchmark growth rate.

We check the calculation at the median scaled NPV, which must be exceeded with probability one half, then vary the target to examine the probability curve.

Updating GBM Parameters with an EMA

An exponential moving average (EMA) gives recent observations more weight. For positive prices spaced $\Delta t > 0$ years apart, use the growth rate $g_k = \ln(S_k/S_{k-1})/\Delta t$. At entry row s , initialize from the training estimates:

$$m_s = \hat{\mu}_{g,0}, \quad v_s = \frac{\hat{\sigma}_0^2}{\Delta t}.$$

Choose a decay factor $0 < \lambda < 1$. For each new observation $k > s$:

$$\begin{aligned}\delta_k &= g_k - m_{k-1}, \\ m_k &= m_{k-1} + (1 - \lambda)\delta_k, \\ v_k &= \lambda[v_{k-1} + (1 - \lambda)\delta_k^2].\end{aligned}$$

Here, m_k and v_k estimate the mean and variance of the growth rate.

EMA: Recover the GBM Parameters

For the growth rate, the estimated mean m_k has units of year^{-1} and the estimated variance v_k has units of year^{-2} . Convert these moments to GBM parameter estimates:

$$\hat{\mu}_{g,k} = m_k, \quad \text{mean growth rate,}$$

$$\hat{\sigma}_k = \sqrt{v_k \Delta t}, \quad \text{volatility,}$$

$$\hat{\mu}_k = \hat{\mu}_{g,k} + \frac{\hat{\sigma}_k^2}{2}, \quad \text{arithmetic drift.}$$

We estimate the volatility parameter by multiplying the growth-rate standard deviation $\sqrt{v_k}$ by $\sqrt{\Delta t}$. The result has units of $\text{year}^{-1/2}$.

Variance update: the recursion accounts for the changing mean; it does not include a finite-sample unbiased correction.

EMA: Interpret the Half-Life

The **half-life** $H_{\text{half}} > 0$ is the number of observations needed to halve the weight of an older observation:

$$\lambda = 2^{-1/H_{\text{half}}}.$$

With a half-life of 21 observations, $\lambda \approx 0.9675$. The updated mean combines 96.75% of the previous mean with 3.25% of the new growth rate.

A shorter half-life makes the estimates respond faster but also makes them more sensitive to noise. The half-life controls the parameter updates; the holding period sets the forecast horizon. Each forecast holds its parameter estimates fixed.

Derivation: [Derive the exponentially weighted moments and GBM parameter updates](#)

Example: Updating the Parameter Estimates

Example: Compare frozen and updated estimates

Can giving recent observations more weight improve our forecasts?

Each day, we update the probability that the original trade will exceed its NPV target at the end of the chosen forward window. We compare frozen parameters with EMA estimates of volatility alone or of both mean growth and volatility.

We evaluate all three forecasts against the same observed outcomes to see whether updating improves predictions.

Optional Advanced Material

The L4b examples develop two related questions:

- **Uncertainty in mean growth**: why can an estimate remain uncertain even with thousands of observations? Examine the observation period, sampling frequency, and their effects on target probabilities.
- **Monte Carlo versus the closed form**: how many paths are needed to estimate a probability accurately? Compare simulation error with the analytical result and introduce variance reduction techniques.

Examples: Mean-growth uncertainty; Monte Carlo target probabilities.

Summary

We connected GBM price predictions with trade probabilities and parameter updates.

Key takeaways:

- **Out-of-sample predictions:** we used historical estimates to describe later prices. Prediction bands apply to individual dates.
- **NPV target probabilities:** we derived the chance of exceeding a target scaled NPV at a scheduled sale for a chosen benchmark growth rate.
- **EMA parameter updates:** we updated mean growth and volatility from growth-rate moments. The half-life controls how quickly older observations lose weight; each forecast holds its estimates fixed.

Next: L5b extends the price model to correlated assets and portfolio weights.